

MSc(IT) Semester III [Autumn 2024-25]

IT644: Web Services & SOA

02-Sep-2024

Duration: 1 hour

Max. Marks: 40

Roll Number _____ Name _____

1. Define: Web Services with two examples.

(2 Marks)

Ans:

2. List standard protocols to implement SOA.

(2 Marks)

Ans:

3. Why abstraction and granularity principles are key to any SOA implementation?

(2 Marks)

Ans:

4. What challenges are associated with quality attributes for web services?

(2 Marks)

Ans:

5. What are the benefits of SOA over monolithic architectures in context of web services? (2 Marks)

Ans:

6. Summarise measures to approach performance requirements. (2 marks)

Ans:

7. What are the key limitations in implementing SOA?

(2 Marks)

Ans:

8. List any four popular software architecture frameworks.

(2 Marks)

Ans:

9. Differentiate: Service-oriented vs Micro-services Architectures.

(2 Marks)

Ans:

10. Define: communication cohesion and content coupling.

(2 marks)

Ans:

11. How are functional and non-functional requirements different?

(2 Marks)

Ans:

12. Differentiate: Reference Architecture vs. Product-line Architecture

(2 marks)

Ans:

13. How to approach portability requirements?

(2 marks)

Ans:

14. How are enterprise, segment and solution architectures related?

(2 Marks)

Ans:

15. Depict key components of SOA using a suitable diagram.

(3 marks)

Ans:

16. Depict commonly considered key NFRs using a suitable diagram.

(3 marks)

Ans:

17. Depict scenarios in software architectural design using a suitable diagram.

(3 marks)

Ans:

18. Using a suitable diagram, show related QAs as per ISO/IEC 25010.

(3 marks)

Ans: